RESEARCH PROGRESS
PRESENTATION 1

AI-POWERED RESEARCH GAP IDENTIFIER & TREND ANALYZER

R26-SE-023

RESEARCH BACKGROUND

Problem: Millions of papers make manual gap detection impractical, leading to unintentional duplication.

Effect: Waste of research funding and institutional resources due to redundant development.

Solution: designing a platform that automates the research process using AI driven approach.

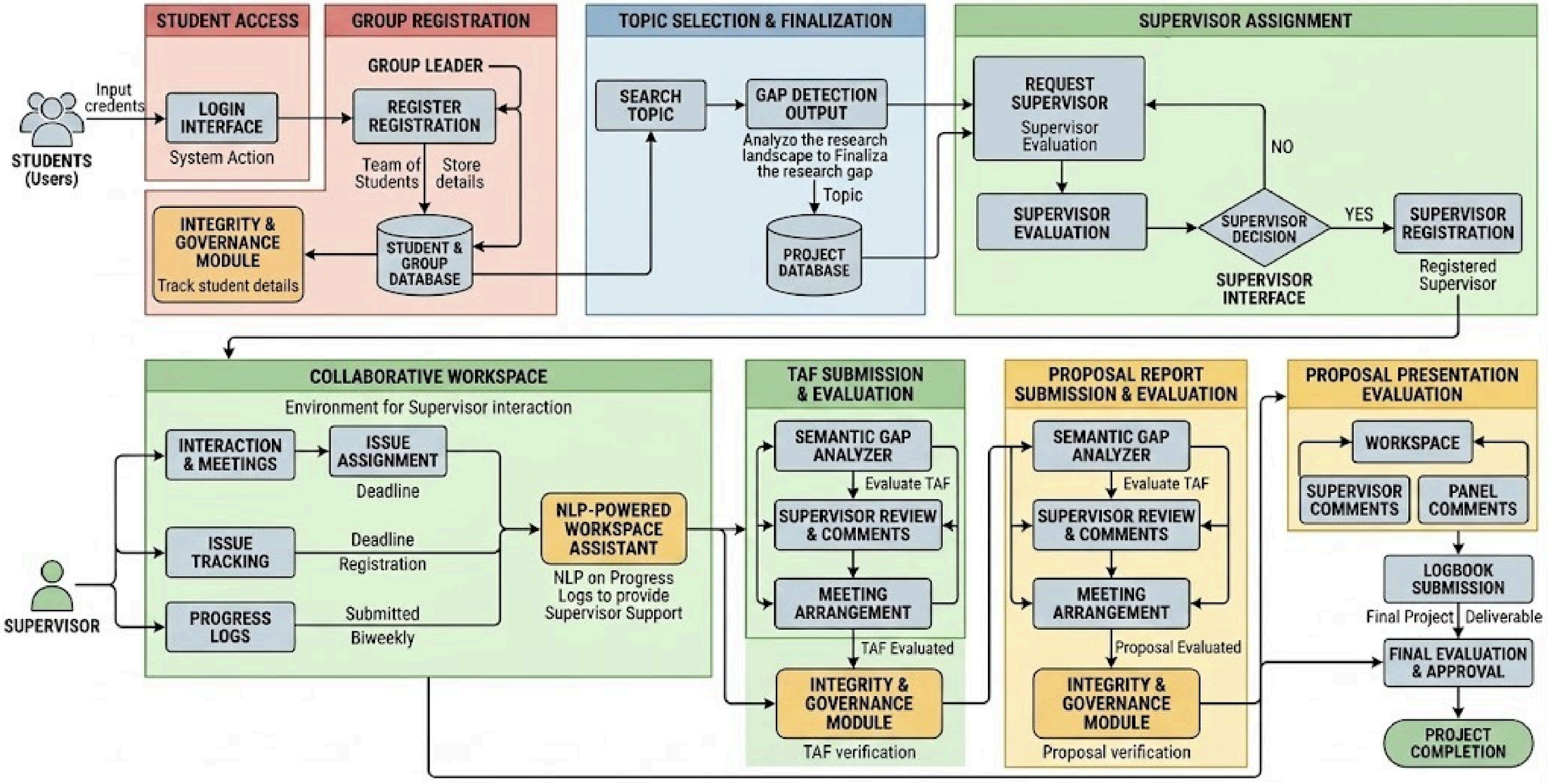
Beneficiaries:

- **Students:** Early validation of novelty.
- **Supervisors:** Quality assurance and automated oversight.
- **Institutions:** Protection of academic reputation and research standards.

COMPONENTS

- Recommendation & Collaboration
- Similarity Analysis & Gap Detection
- Integrity & Governance
- Collaborative Workspace & NPL Powered Supervisor Assistant

Flow Diagram



RECOMMENDATION AND COLLABORATION



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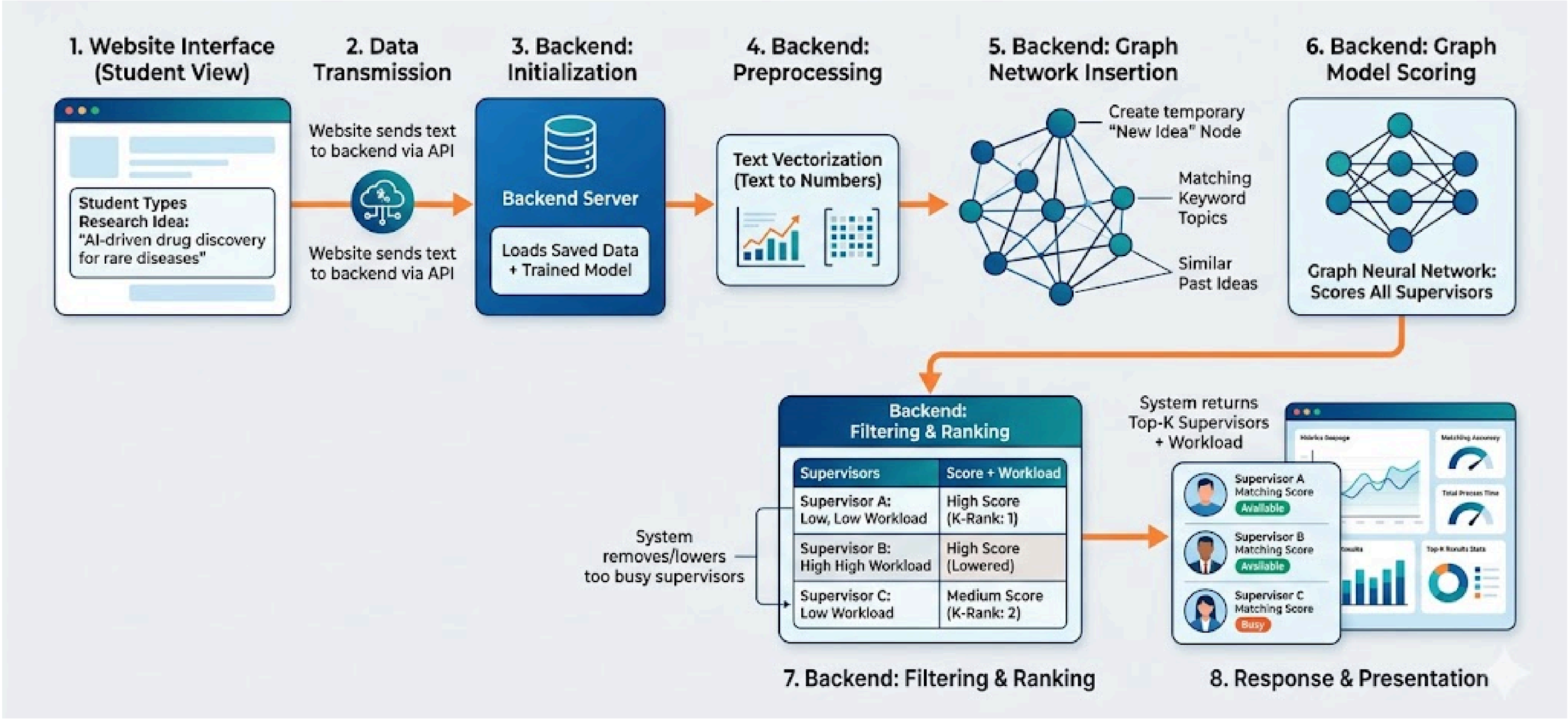
Problem and Solution

- **Problem** - Finding the most suitable and **available** supervisor for a given research idea

- **Solution** - Using Graph Neural network to make a connection between each supervisor and the topic with their availability and suggest the best possible supervisors

- **Novelty -** Using **graphSAGE** with **GNN** instead of TF-IDF or neural network to filter out the best supervisors

Current workflow



Design Excellence

- Uses a heterogeneous graph (idea, supervisor, keyword) to represent the academic environment realistically.
- Uses inductive **GraphSAGE** so the system can handle new ideas without retraining each time by connecting the new idea to keyword neighborhood
- Adds **workload-aware** ranking after GNN scoring

Why using GNN and graphSAGE

- **GNN:** supervisor selection depends on relationships, not just keywords. GNN learns from that network structure.TF-IDF skips relationships and neural network need data as a table which result in losing information
- **GraphSAGE:** it’s inductive, so it can handle new ideas by using their features with neighbors, without retraining every time.

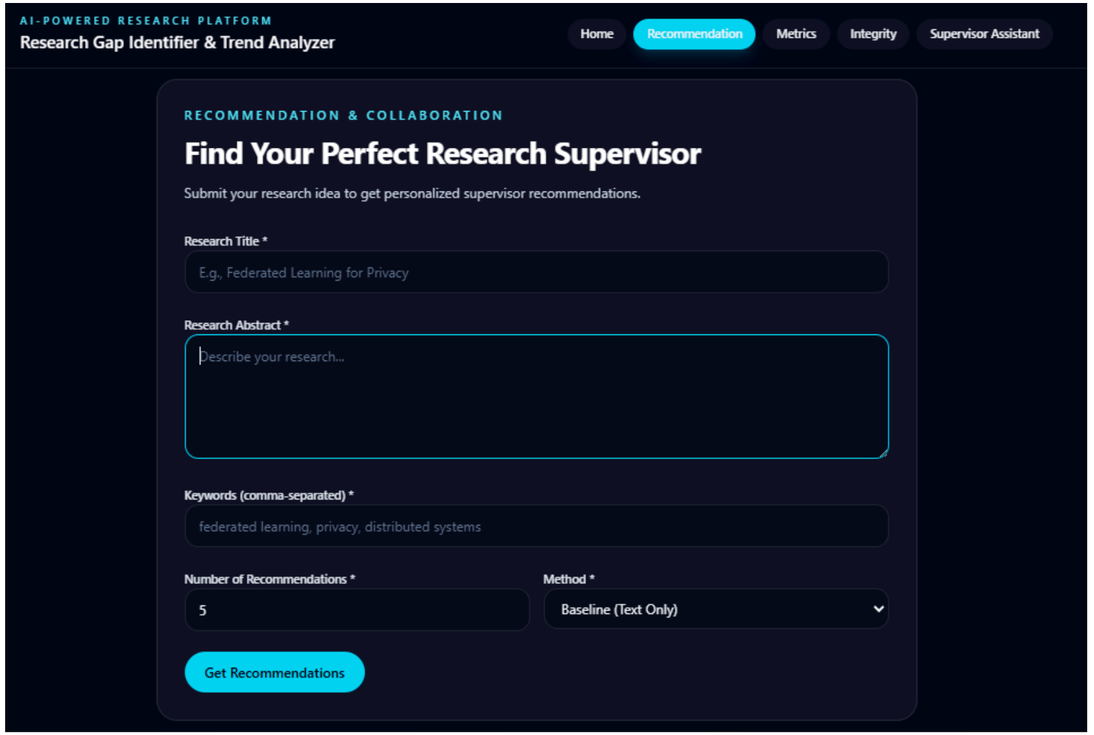
What has been completed

- Heterogeneous graph built using idea–supervisor–keyword nodes + keyword edges
- GraphSAGE GNN training and inference (link prediction)
- Workload filtering after ranking and workload badges in UI

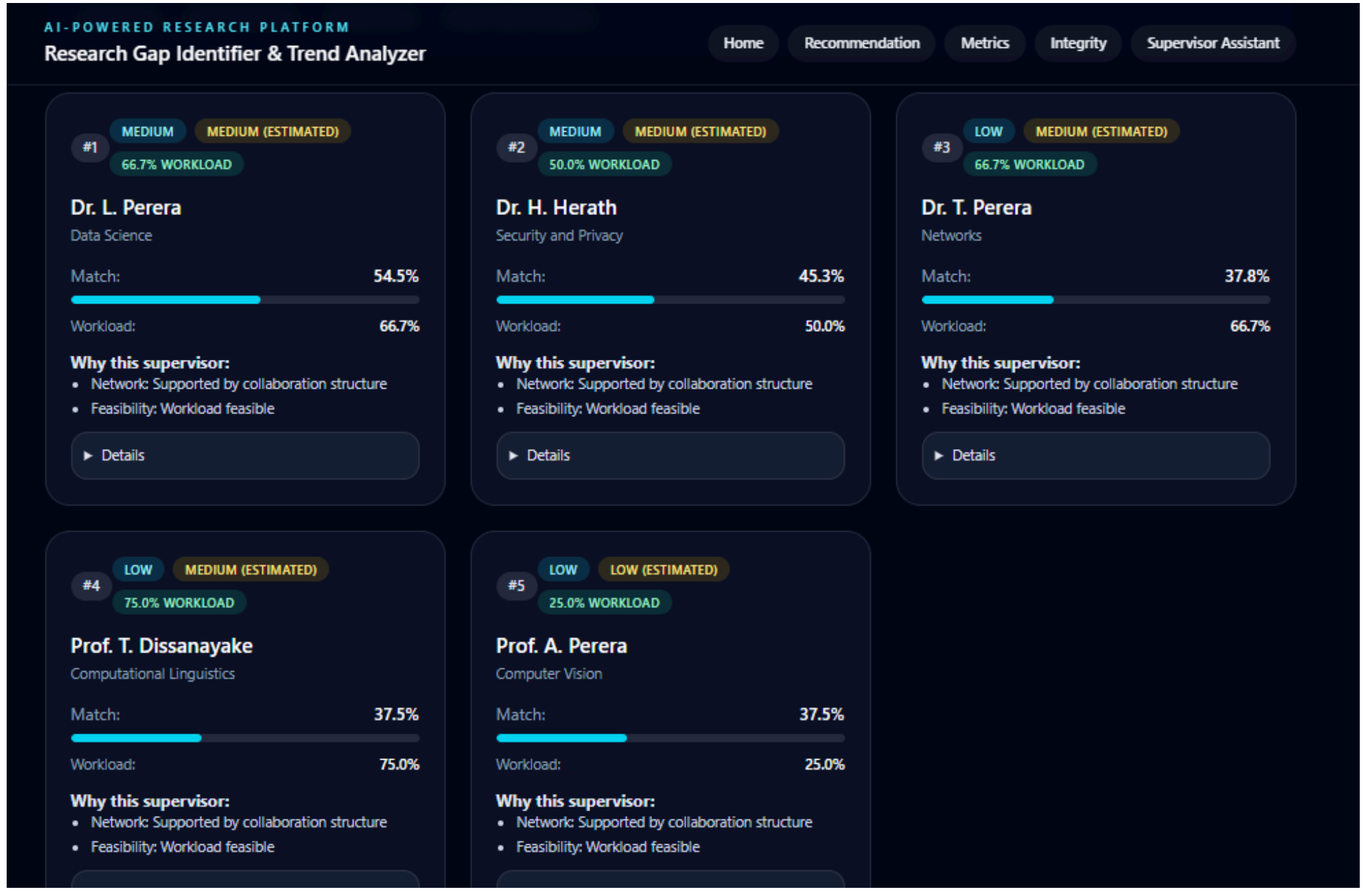
What has to be completed

- supervisor and student registration and workflow
- Database integration (planning with postgreSQL)
- integration with other component

INPUT



OUTPUT



Comparison with other options


```
{
  "baseline_text_only": {
    "hit_at_5": 0.085,
    "precision_at_5": 0.017000000000000005,
    "overload_rate_at_5": 0.20000000000000015,
    "queries": 200
  },
  "baseline_workload": {
    "hit_at_5": 0.045,
    "precision_at_5": 0.009,
    "overload_rate_at_5": 0.0,
    "queries": 200
  },
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    "overload_rate_at_5": 0.0,
    "queries": 200
  }
}
```

SIMILARITY ANALYSIS & GAP DETECTION



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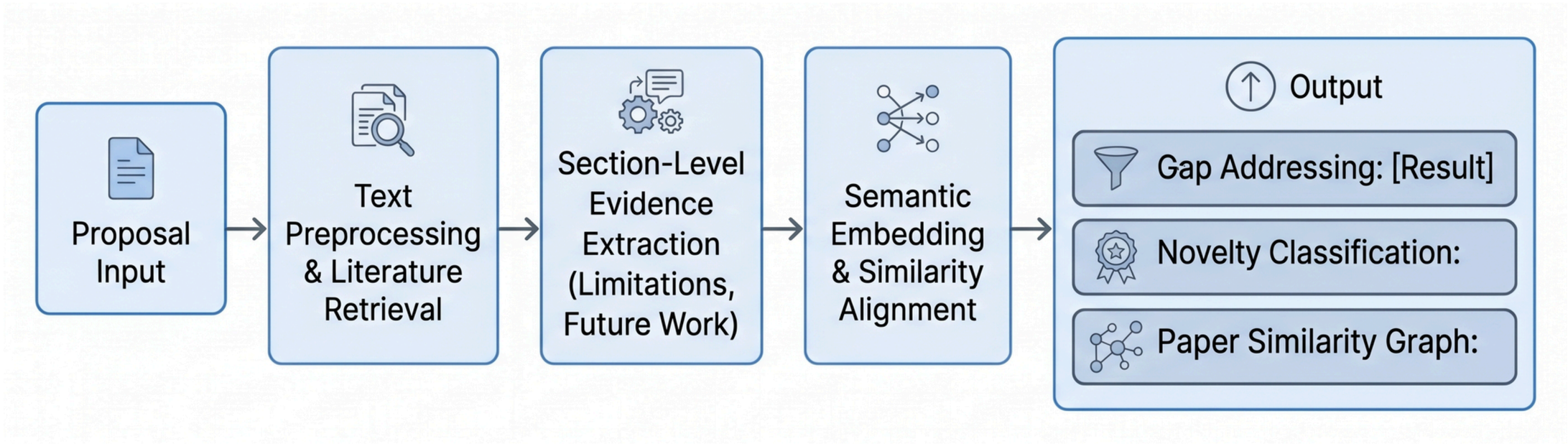
Problem and Solution

Problem - Finding a research idea is slow, inconsistent, and difficult to verify whether a study addresses a genuine research gap or overlaps with existing work.

Solution - A semantic pipeline that automates gap detection and novelty classification by comparing **ideas**/ **proposals** against a exisiting research papers.

Novelty - Instead of relying only on keyword matching, the module combines **similarity scoring** with **LLM reasoning**. This gives clearer, explainable gap classifications and novelty decisions supported by evidence.

Current Workflow



Design Excellence

- Stage-based modular architecture: **Retrieval**, **Extraction**, **Embedding**, **Alignment**, **Reasoning**
- Evidence-first design: every classification is supported by extracted passages
- Extensible output with novelty suggestions and papers similarity graph

Why semantic embeddings instead of TF-IDF

- Embeddings capture **meaning** and **context**, not only exact word overlap.
- Better handling of **paraphrasing** and **domain-specific language**.
- More reliable for identifying **conceptual overlap** and **hidden novelty**.

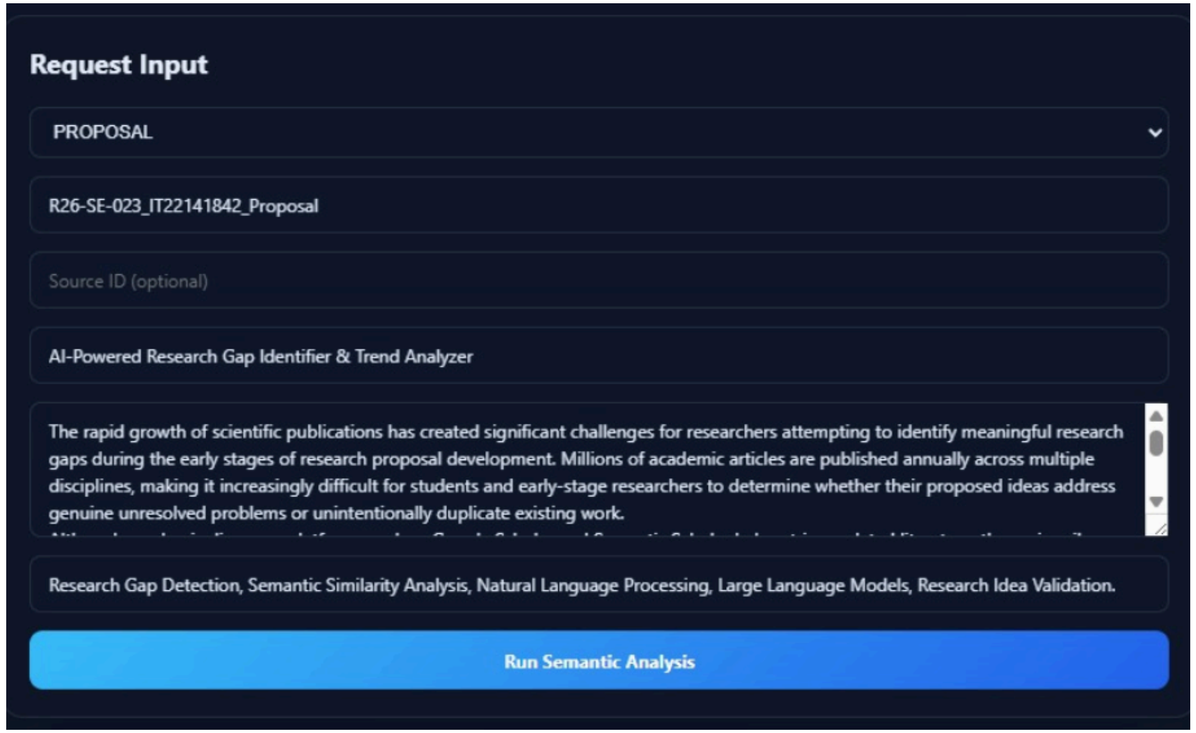
What has been completed

- Core **pipeline** implemented (Retrieval → Extraction → Embedding → Alignment → Reasoning).
- Section-aware evidence extraction with proper fallbacks.
- Embedding + similarity ranking + **LLM-based classification** with **confidence score**.
- **Novelty suggestions** and **paper similarity graph** (connected-papers).

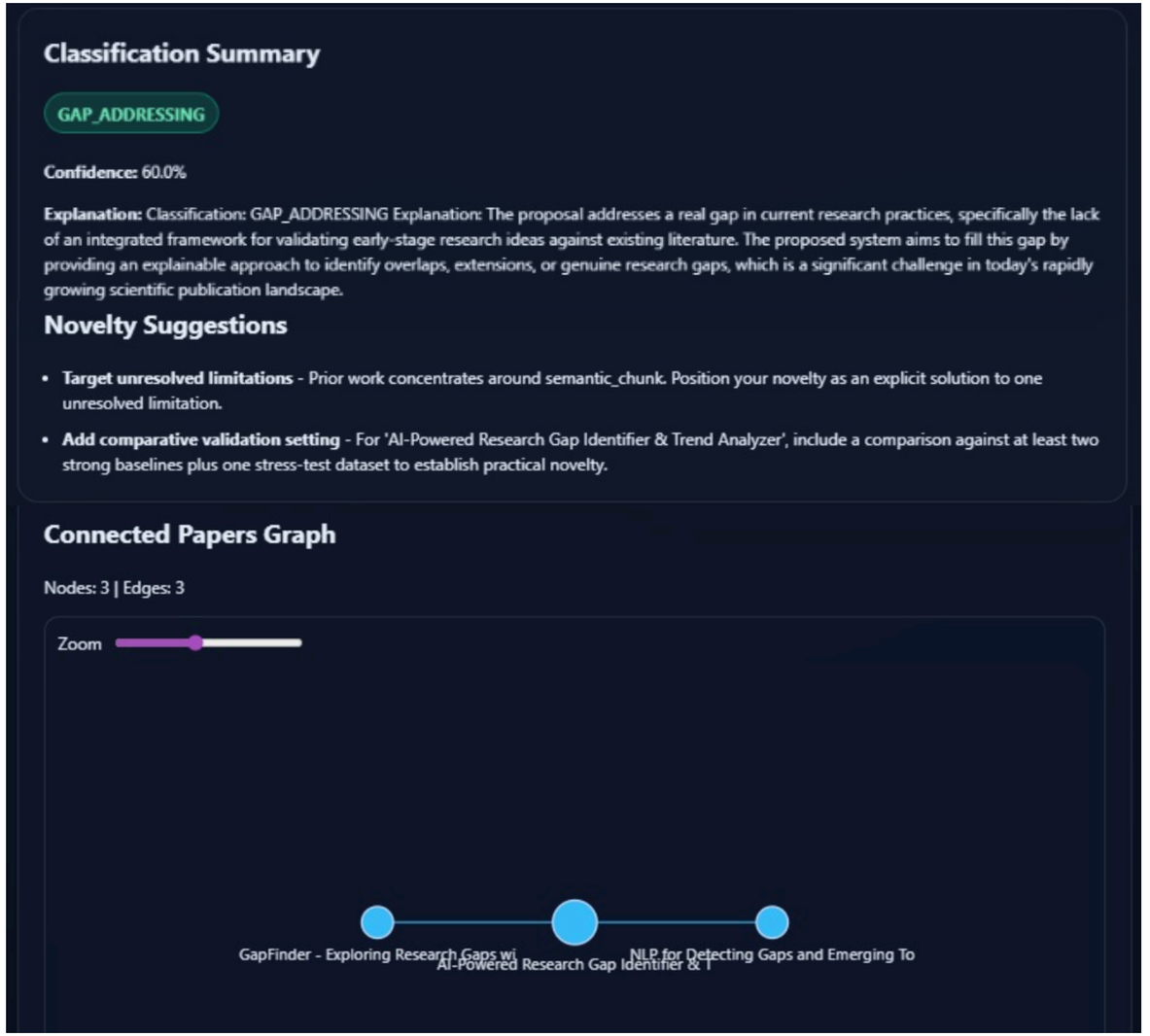
What has to be completed

- **Improve** reasoning, gap summary, and novelty classification.
- Implement **multi stage** gap detection (Topic Ideation, TAF, Proposal).
- Full **integration** with other components and **Database** integration.

INPUT



OUTPUT



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INTEGRITY & GOVERNANCE



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INTEGRITY & GOVERNANCE

Problem and solution

Problem - Existing research systems cannot properly detect similar research topics or securely track research activities.

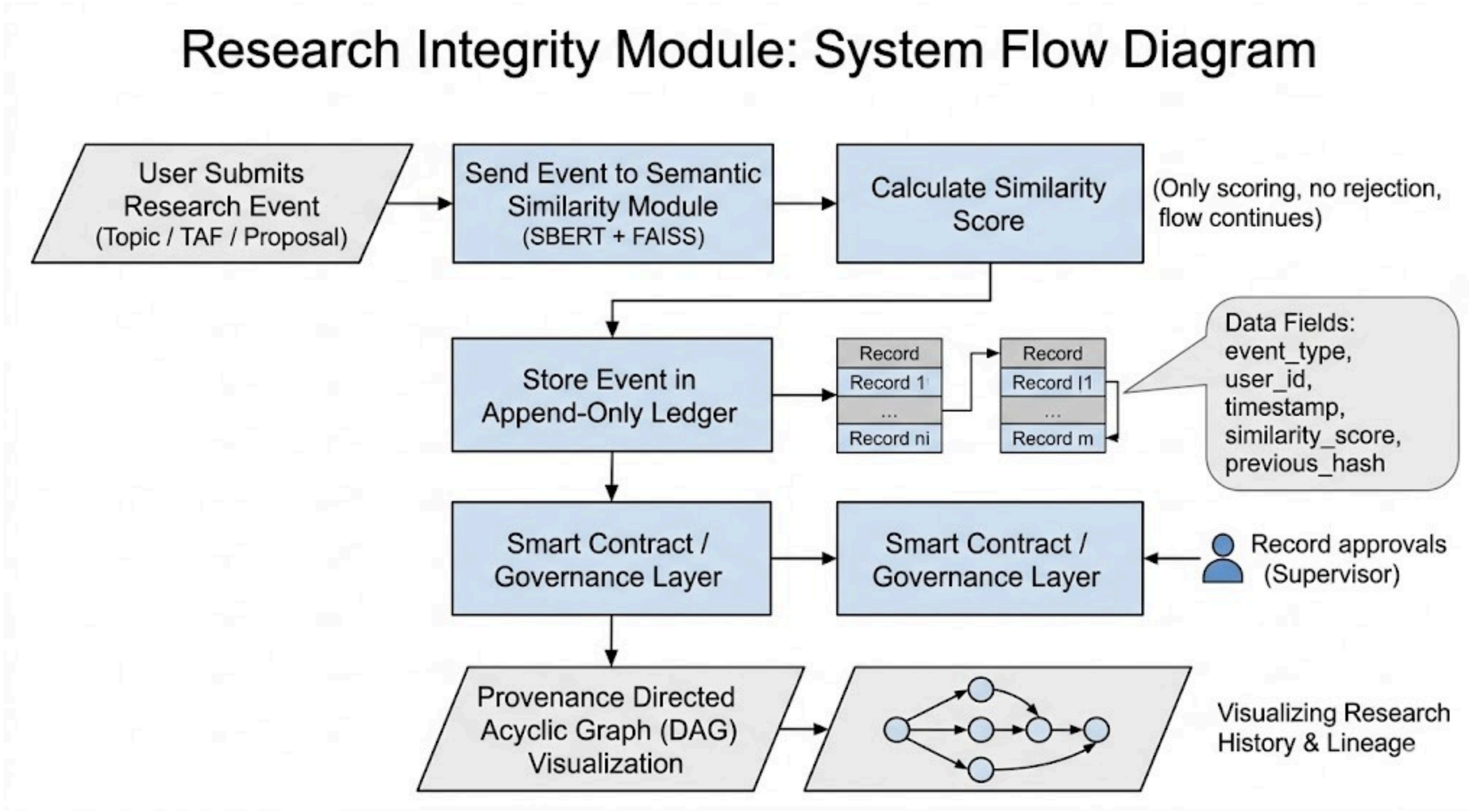
Solution - A blockchain-inspired system using **SBERT + FAISS** and a **hash-linked** ledger to ensure research integrity and traceability.

Novelty -

- Performs real-time semantic similarity analysis using **SBERT + FAISS** to identify overlapping research topics.
- Maintains a **tamper-evident append-only ledger** to track submissions, approvals, and research activity throughout the workflow.

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Workflow - Research event submission → SBERT embedding generation → FAISS similarity search → Similarity score generation → SHA-256 hash-linked ledger storage



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INTEGRITY & GOVERNANCE

Design excellence

- Real-time semantic similarity scoring integrated into research workflow
- Lightweight blockchain-inspired append-only ledger
- Hybrid architecture combining AI semantic analysis + integrity verification
- Modular design for integration with the main Gap Analyzer system

Why SBERT + FAISS?

- SBERT detects semantically similar research topics beyond exact keyword matching.
- FAISS enables fast similarity search across large research datasets.

Why Blockchain-Inspired Instead of Ethereum?

- **A lightweight hash-linked ledger** was used for tamper detection and provenance tracking without full Ethereum complexity and gas fees.

INTEGRITY & GOVERNANCE

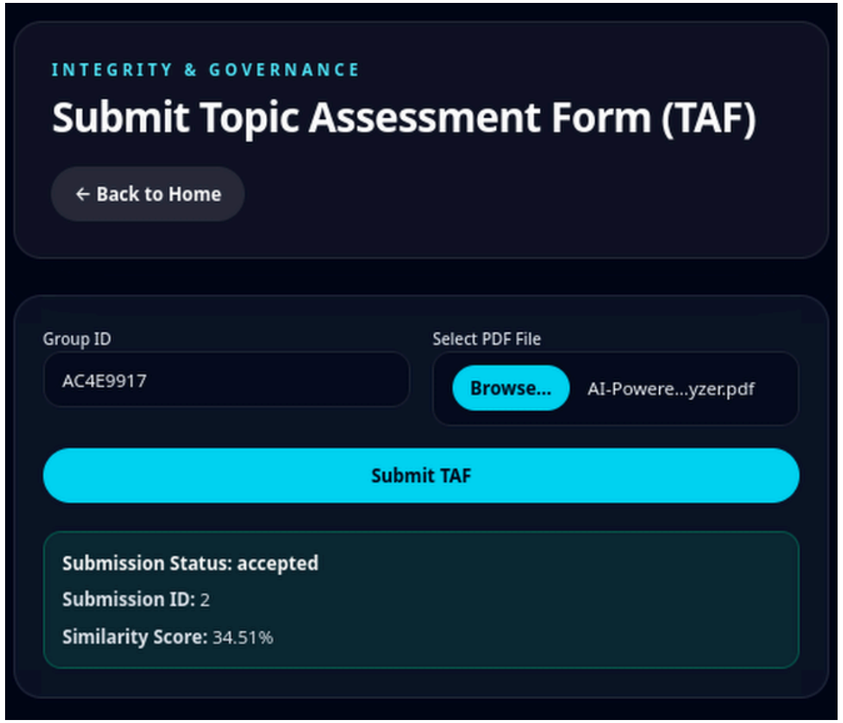
What has been completed

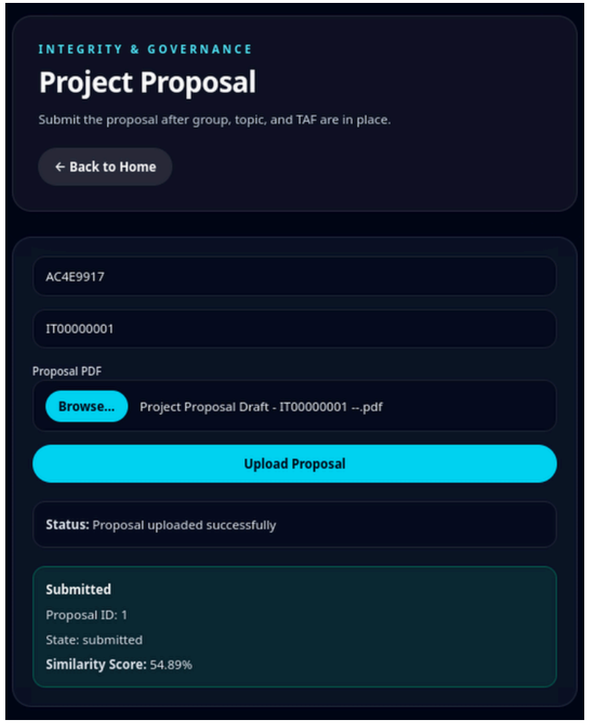
- Research workflow design for submissions and approvals
- **SBERT** embeddings with basic **FAISS** similarity search
- **SHA-256** hash-linked append-only ledger
- Tamper detection for provenance verification
- Initial integration of semantic analysis and integrity tracking
- Dummy research event simulation for testing

What has to be completed

- **PostgreSQL** integration and full system workflow connection
- Role-based tracking for students, supervisors, and evaluators
- **DAG-based** provenance visualization dashboard
- **Smart contract** integration and performance optimization

Semantic Similarity





Tamper detection

Before Tampering

```
~ on  master
❯ curl http://localhost:8001/api/ledger/verify
{"is_valid":true,"block_count":9}⏎
```

After Tampering

```
~ on  master
❯ curl http://localhost:8001/api/ledger/verify
{"is_valid":false,"block_count":9}⏎
```

Tampering DB by manually editing pariticular value

```
sqlite> UPDATE taf_submissions
SET original_filename = 'tampered.pdf'
WHERE id = 1;
sqlite> SELECT * FROM taf_submissions;
```

id	project_group_id	original_filename	stored_path	file_sha256	file_size_bytes	mime_type	uploaded_at	status	similarity_score	top_match_submiss...
1	754D1A66	tampered.pdf	uploads/taf/754D1A66/c3bf3a50a7460b1b048ee77ca434f0f0c91523b2fcb28eebe88c9cf7d327fdf5.pdf	c3bf3a50a7460b1b048ee77ca434f0f0c91523b2fcb28eebe88c9cf7d327fdf5	618179	application/pdf	1778269722.9165442	accepted	NULL	NULL
2	AC4E9917	AI-Powered Research Gap Identifier & Trend Analyzer.pdf	uploads/taf/AC4E9917/cebcd9ba4e4a814e6baabcc6172a5c13a81c30a1473e764e558ac90a15c28eab.pdf	cebcd9ba4e4a814e6baabcc6172a5c13a81c30a1473e764e558ac90a15c28eab	474403	application/pdf	1778270183.9573016	accepted	0.34506788849830627	NULL

```
sqlite>
```

COLLABORATIVE WORKSPACE & NLP POWERED SUPERVISOR ASSISTANT



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COLLABORATIVE WORKSPACE & NLP POWERED SUPERVISOR ASSISTANT

Problem and solution

Problem - Supervisors spend significant time **manually reviewing research logbooks**, while students lack a centralized system for research management and timely feedback.

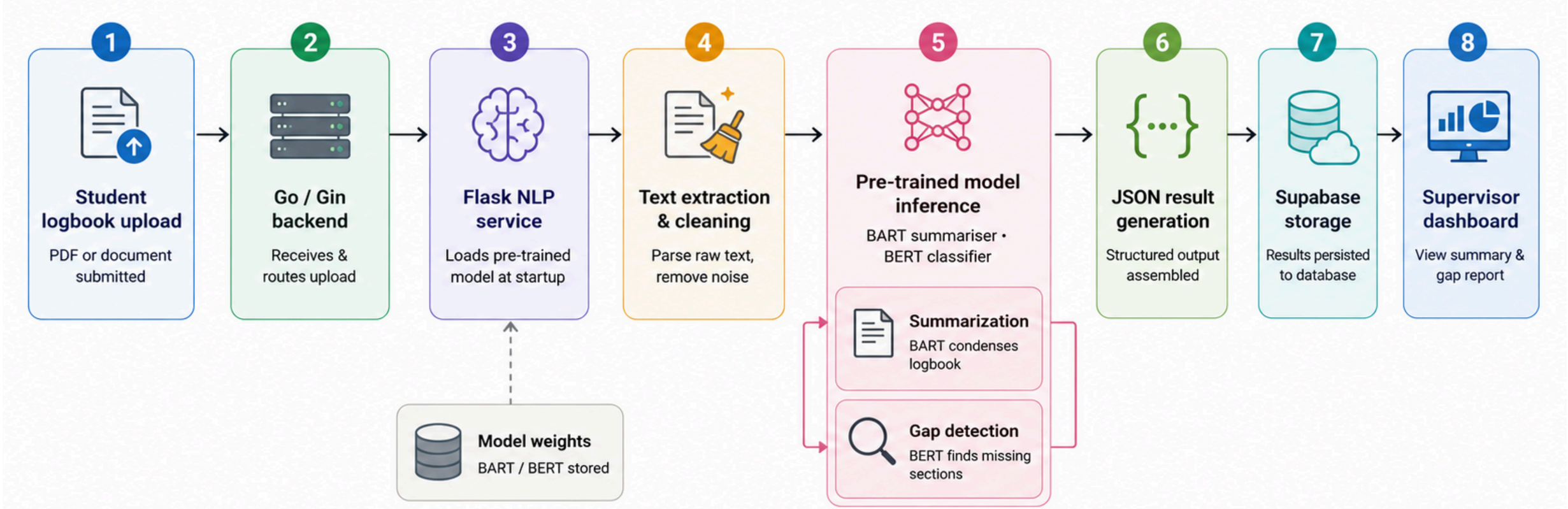
Solution - A collaborative research platform that uses **NLP** to automatically analyze student logbooks, generate progress summaries, and detect missing sections, reducing supervisors' manual workload.

Novelty - Unlike general collaboration platforms such as Slack, this system integrates a **multi-model NLP pipeline** to automatically evaluate research progress quality, summarize logbooks, and support supervisor–student research collaboration within a single workspace.

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COLLABORATIVE WORKSPACE & NLP POWERED SUPERVISOR ASSISTANT

Current workflow



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Design excellence

- Modular **NLP pipeline** with separate components for summarization, gap detection, and confusion analysis
- Scalable Flask microservice architecture enabling independent deployment and seamless frontend integration

Why BART and BERT instead of rule-based or keyword methods

- Transformer models **(BERT, BART)** understand context and meaning, allowing detection of implicit gaps and confusion beyond rule-based keywords.
- They support abstractive summarization and zero-shot classification, making the system more accurate, flexible, and scalable across research domains.

COLLABORATIVE WORKSPACE & NLP POWERED SUPERVISOR ASSISTANT

What has been completed

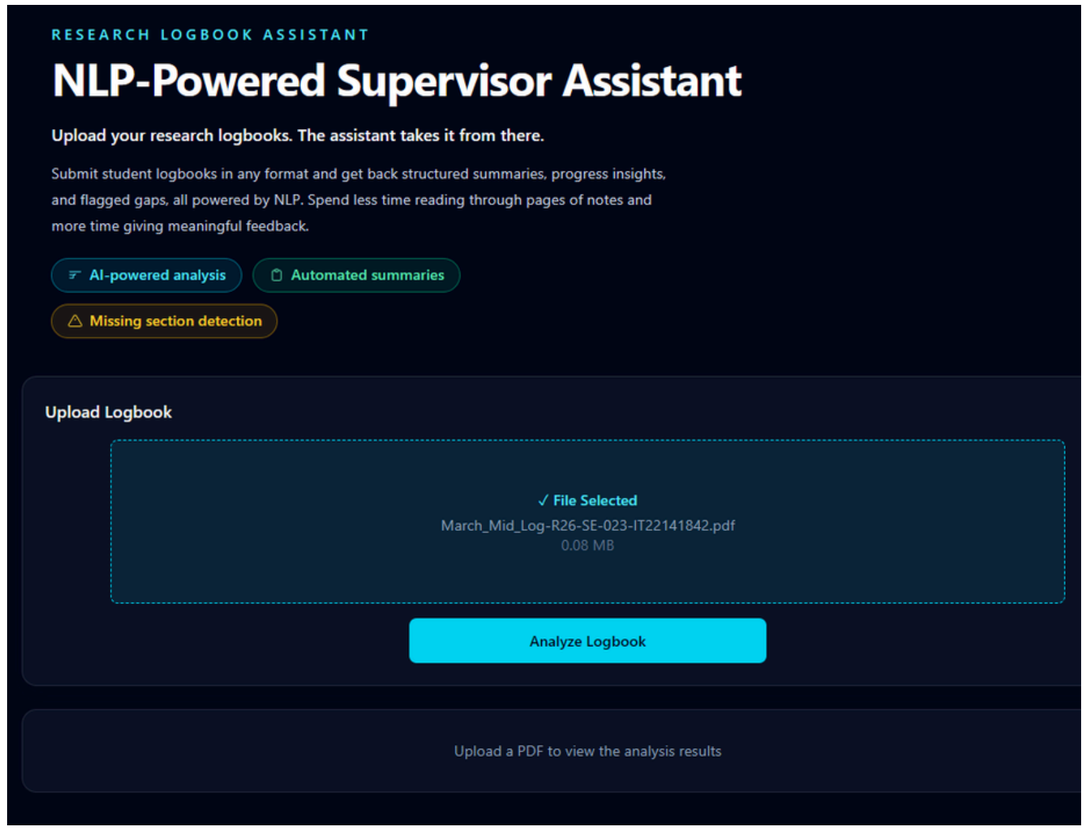
- Frontend login flow implemented
- Student dashboard page developed
- Pre-trained model integration (BART for summarization, BERT for gap detection) loaded into Flask NLP service
- Figma prototypes designed for student and supervisor workspace

What has to be completed

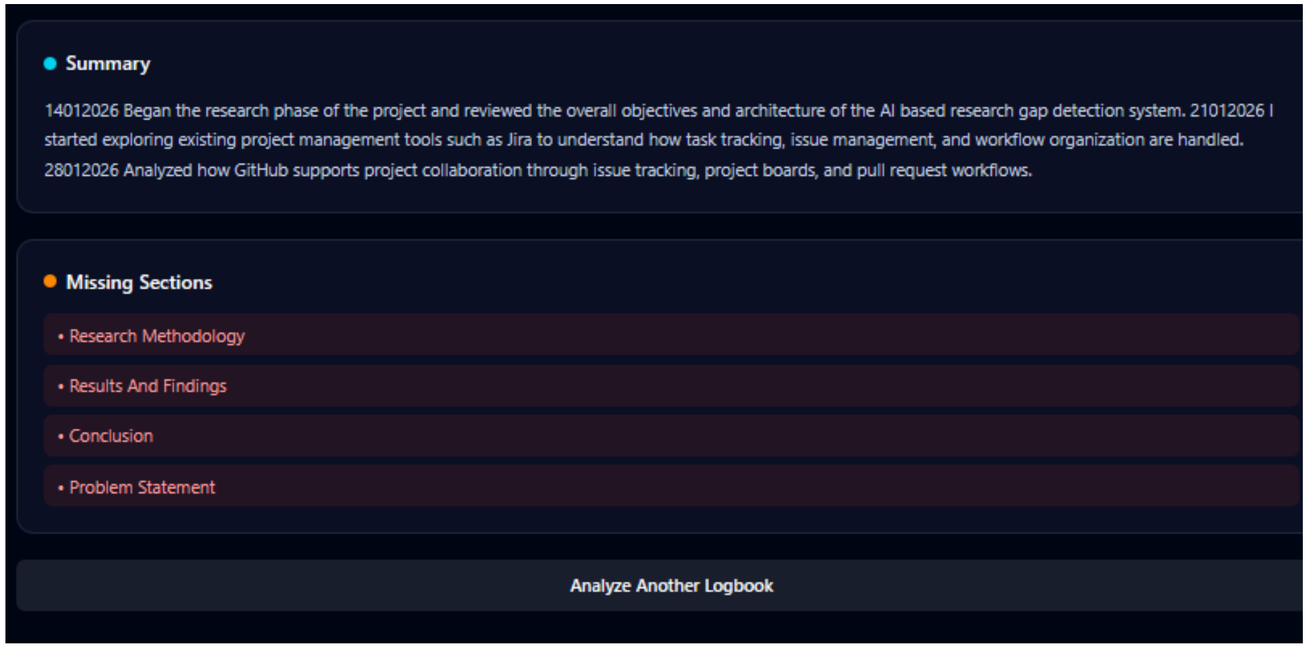
- NLP model fine-tuning on research logbook domain data
- Model accuracy evaluation (precision, recall, F1-score)
- Supervisor dashboard pages development
- End-to-end integration testing (upload → NLP → dashboard)
- Deployment and performance optimization

COLLABORATIVE WORKSPACE & NLP POWERED SUPERVISOR ASSISTANT

INPUT



OUTPUT



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THANK YOU

QUESTIONS & DISCUSSION